Use Case: Place Piece

Iteration: 1

Primary Actor: Player

Goal in Context: Allow the player to place a game piece on the board during their turn.

Preconditions:

1. The player is logged into their account.
2. The player is connected to an active game session.
3. It is the player's turn to make a move.
4. The player has selected a specific grid location on the game board.

Trigger: The player decides to place a piece on the game board during their turn.

Scenario:

1. The player selects an available spot on the game board by clicking or tapping.
2. The system does a verification check that it is the player's turn.
3. The system checks if the move is valid.
4. The system processes the piece placement based on the game type:
    a. Tic-Tac-Toe
        i. The player selects a square on the 3x3 grid.
    b. Connect 4
        i. The player selects a column in the grid.
        ii. The system drops the piece to the lowest available row in that column.
    c. Checkers
        i. The player selects a valid move for their checker piece, following standard movement rules.
        ii. The system verifies for a capture
5. The system updates the game state and checks for any winning or game-ending conditions.
6. The system notifies the opponent that it is now their turn.

Post Conditions:

1. The game board is updated with the newly placed piece.
2. If a winning condition is met, the game concludes and a winner is declared.
3. Else the turn is handed over to the opponent.

Exceptions:

- The player attempts to place a piece outside the boundaries of the board.
- The player selects an occupied spot (e.g., in Tic-Tac-Toe).
- The player makes an invalid move for the selected piece (e.g., in Checkers, pieces can only move diagonally on dark squares).

Priority: High, as placing a piece is a core action in each game.

When available: 1st iteration

Frequency of Use: Multiple times per game, depending on game length.

Channel to Actor: The player interacts through the screen by tapping or clicking on the board.

Secondary Actors: Opponent player.

Channels to Secondary Actors: Updates appear on the opponent's screen to reflect the placed piece and turn change.

Open Issues: None.